

C++ Programming Language

Comprehensive Theory, Core Concepts, OOP Examples & 25 Practice Questions

1. Introduction to C++ Programming

C++ is a high-level, general-purpose programming language developed by Bjarne Stroustrup at Bell Labs in 1979 as an extension of the C language. It combines procedural, object-oriented (OOP), and generic programming features, making it ideal for system software, game development, high-performance engines, and real-time applications.

Key Features of C++ Language

- **Object-Oriented Programming (OOP):** Supports Classes, Objects, Inheritance, Polymorphism, Encapsulation, and Abstraction.
- **Rich Standard Template Library (STL):** Provides ready-to-use data structures (vectors, maps) and algorithms.
- **Memory Control:** Offers direct pointer manipulation alongside dynamic allocation (`new` / `delete`).
- **Multi-Paradigm:** Allows writing procedural, functional, and object-oriented code in the same project.

2. Core C++ Concepts & Code Examples

1. Basic Structure, Input & Output

C++ uses `std::cout` for printing and `std::cin` for receiving input. Header file `<iostream>` is required.

```
#include <iostream>
using namespace std;

int main() {
    int age;
    cout << "Enter your age: ";
    cin >> age;
    cout << "You are " << age << " years old." << endl;
    return 0;
}
```

2. Classes & Objects (Encapsulation)

Classes wrap data (attributes) and methods (functions) together. Access specifiers (public, private, protected) enforce encapsulation.

```
#include <iostream>
using namespace std;

class Student {
private:
    string name;
public:
    void setName(string n) { name = n; }
    void display() { cout << "Name: " << name << endl; }
};

int main() {
    Student s;
    s.setName("Rahul");
    s.display();
    return 0;
}
```

3. Inheritance

Inheritance allows a derived class to inherit properties and behavior from a base class, promoting code reusability.

```
#include <iostream>
using namespace std;

class Animal {
public:
    void eat() { cout << "Eating..." << endl; }
};

class Dog : public Animal {
public:
    void bark() { cout << "Barking..." << endl; }
};

int main() {
    Dog d;
    d.eat();
    d.bark();
    return 0;
}
```

4. Polymorphism & Virtual Functions

Polymorphism allows functions to perform different actions based on the object invoking them (Runtime polymorphism via virtual functions).

```
#include <iostream>
using namespace std;

class Base {
public:
    virtual void show() { cout << "Base Show" << endl; }
};

class Derived : public Base {
public:
    void show() override { cout << "Derived Show" << endl; }
};

int main() {
    Base* b = new Derived();
    b->show(); // Outputs: Derived Show
    delete b;
    return 0;
}
```

5. Standard Template Library (STL Vector)

Vectors are dynamic arrays that automatically resize when elements are added or removed.

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> nums = {10, 20, 30};
    nums.push_back(40);

    for (int n : nums) {
        cout << n << " ";
    }
    return 0;
}
```

3. Constructors, Exception Handling & Templates

- **Constructors & Destructors:** Special member functions called automatically when an object is created or destroyed.
- **Function & Class Templates:** Enable generic programming without binding to specific data types.
- **Exception Handling:** Uses try, catch, and throw blocks to handle runtime errors gracefully.
- **Dynamic Memory (`new` / `delete`):** Operators used for allocating and freeing memory on the heap.

4. C++ Language Practice Questions (25 MCQs)

1. Who developed the C++ programming language?

- A) Dennis Ritchie
- B) Bjarne Stroustrup
- C) James Gosling
- D) Guido van Rossum

2. Which concept in C++ allows creating a new class from an existing class?

- A) Encapsulation
- B) Inheritance
- C) Abstraction
- D) Polymorphism

3. Which header file is used for basic input and output streams in C++?

- A) <stdio.h>
- B) <iostream>
- C) <fstream>
- D) <stdlib.h>

4. Which operator is used for dynamic memory allocation in C++?

- A) malloc
- B) new
- C) alloc
- D) create

5. Which keyword is used to deallocate memory allocated with 'new'?

- A) free
- B) delete
- C) remove
- D) destruct

6. Which access specifier makes members accessible only within the same class?

- A) public
- B) private
- C) protected
- D) internal

7. What is a constructor in C++?

- A) A method to delete objects
- B) A special member function to initialize objects
- C) A keyword for inheritance
- D) A standard library function

8. Which symbol is used to define a destructor in C++?

- A) !
- B) ~
- C) #
- D) *

9. What feature enables runtime polymorphism in C++?

- A) Inline functions
- B) Virtual functions
- C) Friend functions
- D) Static functions

10. What stands for STL in C++?

- A) Standard Text Library
- B) Standard Template Library
- C) System Type Library
- D) Structured Tool Library

11. Which operator is called the scope resolution operator in C++?

- A) ->
- B) ::
- C) .
- D) :

12. Can a constructor be overloaded in C++?

- A) No
- B) Yes
- C) Only copy constructor
- D) Only default constructor

13. Which keyword is used to handle exceptions in C++?

- A) try / catch
- B) do / catch
- C) handle / catch
- D) throw / handle

14. What type of inheritance occurs when a class inherits from multiple base classes?

- A) Single Inheritance
- B) Multiple Inheritance
- C) Multilevel Inheritance
- D) Hierarchical Inheritance

15. What is the output of `std::cout << (5 == 5)`?

- A) true
- B) 1
- C) 0
- D) false

16. Which function non-member access grants private/protected access to a class?

- A) Inline function
- B) Friend function
- C) Static function
- D) Virtual function

17. What keyword is used to define generic functions or classes?

- A) generic
- B) template
- C) typename
- D) macro

18. Which container in STL implements dynamic arrays?

- A) list
- B) vector
- C) map
- D) queue

19. What is wrapping up of data and functions into a single unit called?

- A) Abstraction
- B) Encapsulation
- C) Polymorphism
- D) Inheritance

20. What is a class containing at least one pure virtual function called?

- A) Concrete class
- B) Abstract class
- C) Friend class
- D) Base class

21. What is the syntax for a pure virtual function?

- A) `virtual void show() = 0;`
- B) `virtual void show();`
- C) `void show() = pure;`
- D) `virtual void show() = NULL;`

22. Which stream class is used for file output in C++?

- A) ifstream
- B) ofstream
- C) fstream
- D) iostream

23. What keyword prevents a class from being inherited or a method from being overridden?

- A) static
- B) final
- C) const
- D) sealed

24. What does 'this' pointer point to inside a class method?

- A) Base class
- B) Current object instance
- C) Parent object
- D) Global scope

25. Which function is used to get the size of dynamic vector in C++?

- A) length()
- B) size()
- C) count()
- D) capacity()

Answer Key

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|------|-------|-------|-------|-------|
| 1. B | 6. B | 11. B | 16. B | 21. A |
| 2. B | 7. B | 12. B | 17. B | 22. B |
| 3. B | 8. B | 13. A | 18. B | 23. B |
| 4. B | 9. B | 14. B | 19. B | 24. B |
| 5. B | 10. B | 15. B | 20. B | 25. B |